

7 Sweet pea plants were originally found in Sicily and southern Italy.

They are used to investigate genetic inheritance and linkage.

The photograph shows a flowering sweet pea plant.



(Source: © Tim Gainey/Alamy Stock Photo)

Sweet pea plant flowers can be different colours and have a variety of pollen grain shapes.

(a) (i) How many of the following statements about a pollen grain are correct?

- contains a tube nucleus and a generative nucleus
- the generative nucleus divides by mitosis to form two female gametes
- can produce diploid gametes

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

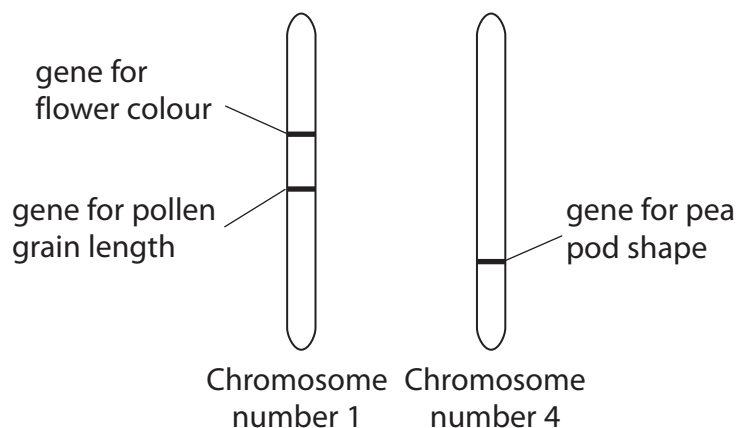




- (c) Sweet pea plants were used in an investigation into the inheritance of flower colour, pollen grain length and pea pod shape.

Each of these traits is controlled by a single gene. Sweet peas have seven pairs of chromosomes.

The arrangement of these three genes on sweet pea chromosomes is shown in the diagram.



Comment on the role of meiosis in the inheritance of these traits.

(4)

(Total for Question 7 = 11 marks)

